

C.5 Doppler effect

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **24 marks**

Speed of sound in air = 340 m s^{-1} ; $c = 3.0 \times 10^8 \text{ m s}^{-1}$. Show your working.

Question 1 — The gelato van

(11 marks)

A gelato van drives down a long straight street at 25 m s^{-1} (the driver is late), playing its jingle at a steady 700 Hz. You stand on the footpath as it approaches, passes, and recedes.

- (a) Describe the Doppler effect for sound, in terms of wavefronts. [2]
- (b) [HL] Calculate the frequency you hear while the van approaches. [3]
- (c) [HL] Calculate the frequency you hear once it has passed. [2]
- (d) Sketch a graph of the frequency you hear against time, from well before the van reaches you until well after. Mark the value 700 Hz on your frequency axis. [3]
- (e) State one use of the Doppler effect in medicine or in weather forecasting. [1]

Question 2 — The learner driver

(6 marks)

The same van, now driven by a learner, sits parked in front of you with the jingle playing, then accelerates *away* smoothly up the street.

- (a) State the frequency you hear while the van is parked, and explain why there is no shift, however loud the jingle. [2]
- (b) Describe how the frequency you hear changes from the moment the van starts moving until it reaches a steady speed. [2]
- (c) A classmate claims “any moving source gives a constant Doppler shift”. Use this scenario to correct the claim. [2]

Question 3 — [HL] The runaway galaxy

(7 marks)

In the lab, a hydrogen emission line has wavelength 656.0 nm. In the spectrum of a distant galaxy, the same line arrives at 659.2 nm.

- (a) State whether the light is red-shifted or blue-shifted, and what this tells you about the galaxy’s motion. [2]
- (b) Calculate the galaxy’s speed along the line of sight. [3]
- (c) Explain why measurements like this one, repeated across many galaxies, matter to cosmology. [2]

Answers

Question 1 — The gelato van

- (a) A moving source partly catches up with the wavefronts it emits ahead of itself, so those fronts are crowded closer together: shorter wavelength and therefore higher observed frequency in front. Behind the source the fronts are drawn further apart, giving longer wavelength and lower frequency. The source emits at an unchanged rate throughout; it is the *spacing* of the fronts reaching the observer that changes.
- (b) $f' = f \frac{v}{v - u_s} = 700 \times \frac{340}{340 - 25} = 700 \times \frac{340}{315} = 756 \text{ Hz}.$
- (c) $f' = f \frac{v}{v + u_s} = 700 \times \frac{340}{365} = 652 \text{ Hz}.$
- (d) Roughly constant at about 756 Hz while approaching, then a rapid fall *through* 700 Hz as the van passes, settling at roughly 652 Hz as it recedes. The graph is a step-like fall, not a gradual slide, and 700 Hz is heard only momentarily, at closest approach.
- (e) Any one of: ultrasound measurement of blood flow; Doppler radar for rain and wind in weather forecasting; foetal heart monitors.

Question 2 — The learner driver

- (a) 700 Hz, unshifted. The source is at rest, so its wavefronts spread concentrically and reach you at exactly the rate they are emitted. Loudness is amplitude, which has nothing to do with frequency.
- (b) It starts at 700 Hz and falls continuously as the van gains speed, because the shift grows with the source's speed. Once the van reaches a steady speed the frequency levels off at the constant receding value.
- (c) The shift depends on the source's *speed*, not merely on whether it is moving. An accelerating source gives a continuously changing shift, and a stationary source gives none at all. A constant shift requires constant velocity.

Question 3 — [HL] The runaway galaxy

- (a) The wavelength has *increased* ($656.0 \rightarrow 659.2 \text{ nm}$), so the light is red-shifted and the galaxy is receding from us.
- (b) $\Delta\lambda = 3.2 \text{ nm}$, so $v = c \frac{\Delta\lambda}{\lambda} = 3.0 \times 10^8 \times \frac{3.2}{656.0} = 1.5 \times 10^6 \text{ m s}^{-1}.$
- (c) Repeated across many galaxies, the measurements show that recession speed increases with distance (Hubble's law). That systematic red shift of distant galaxies is the observational evidence that the universe is expanding.